Obesity Category Prediction Using Machine Learning: A Comparative Study of Supervised and Unsupervised Approaches

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Abstract—Obesity is a multifactorial public-health crisis driven by complex interactions among dietary habits, physical activity, genetic predisposition, and socioeconomic factors. Accurate categorization of individuals into clinically meaningful obesity classes enables targeted interventions and resource allocation. This paper presents a comprehensive machine-learning study on the *Updated Obesity Dataset* (2,111 records, 16 features, 7 classes). Five supervised classifiers—K-Nearest Neighbors (KNN), Decision Tree, Logistic Regression, Naive Bayes, and a Multilayer Perceptron (MLP)—are trained, evaluated, and compared using accuracy, macro-averaged precision, recall, F1-score, and one-vs-rest AUC. Unsupervised KMeans clustering is additionally applied to explore natural data groupings. The MLP achieves the highest test accuracy of 93.21% with an AUC of 0.9884, while KMeans yields a silhouette score of 0.31, confirming moderate natural cluster structure.

Keywords—obesity classification, machine learning, neural network, decision tree, logistic regression, Naive Bayes, KNN, KMeans clustering, feature preprocessing, healthcare AI

I. INTRODUCTION

Obesity is recognized by the World Health Organization as one of the most pressing public-health crises of the twenty-first century. Its global prevalence has more than tripled since 1975, and it is now a leading risk factor for cardiovascular disease, type 2 diabetes, certain cancers, and musculoskeletal disorders [1]. Traditional clinical screening relies predominantly on Body Mass Index (BMI), but BMI alone is insufficient to capture the behavioral, dietary, and lifestyle dimensions that drive obesity progression.

Machine learning offers a powerful complementary approach: by ingesting rich multi-modal feature sets, ML models can discover non-linear decision boundaries that elude linear risk scores. This project—developed as part of the CSE422 Artificial Intelligence Laboratory at Brac University—systematically benchmarks five widely used classifiers on a curated obesity dataset and addresses three core research questions: (1) Which classifier best predicts obesity category from lifestyle features? (2) How do interpretable models compare with neural networks? (3) Do natural clusters in the feature space align with clinical obesity classes?

II. DATASET DESCRIPTION

The *Updated Obesity Dataset* contains **2,111 records** and **17 columns**—16 input features plus one target label. The target column **NObeyesdad** encodes seven distinct obesity categories. As shown in Fig. 1, the class distribution is moderately imbalanced, with Obesity Type I being the most frequent class (351 samples, ~16.6%) and Insufficient Weight the least frequent (148 samples, ~7.0%).

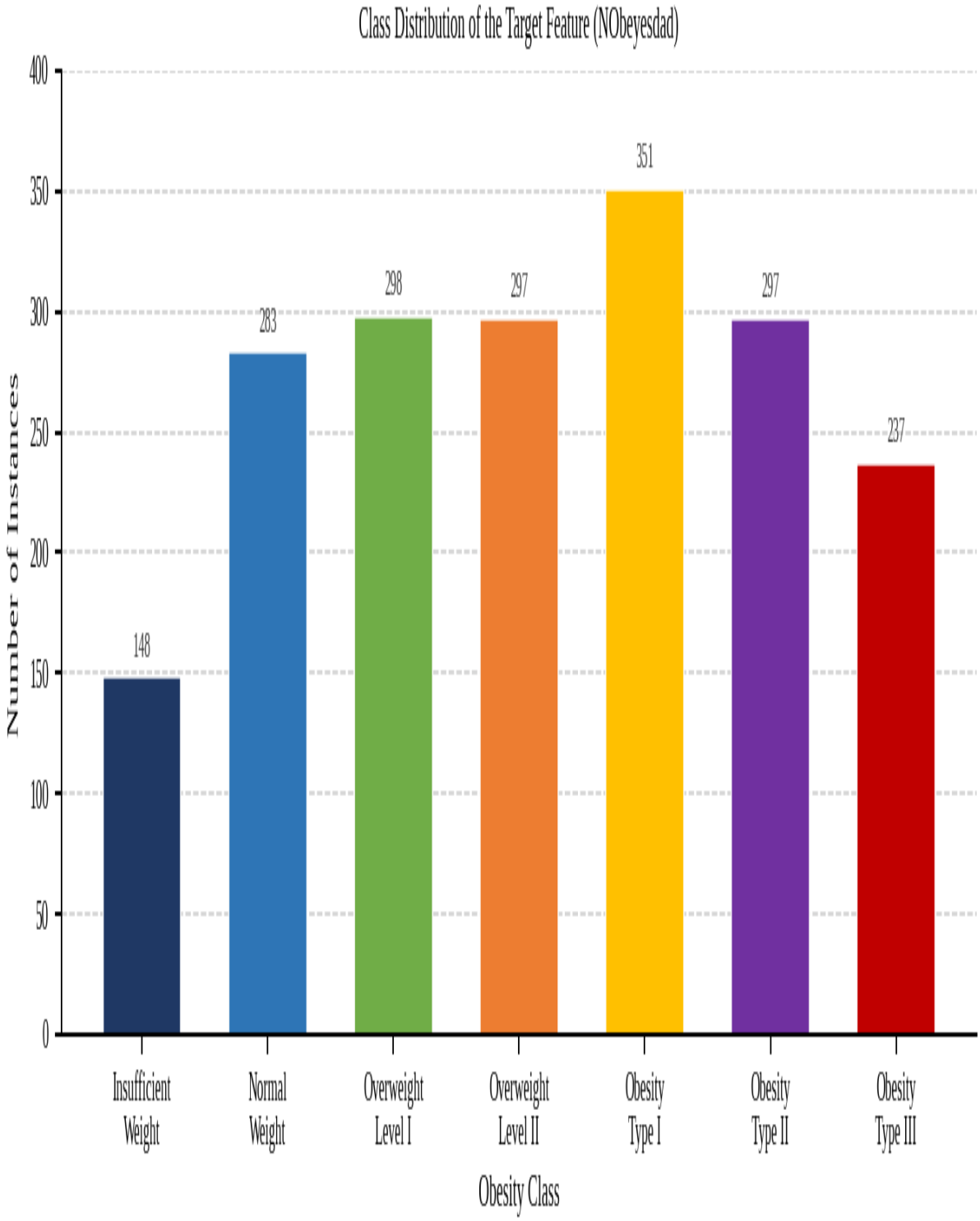

Fig. 1. Class distribution of the target feature NObeyesdad across the seven obesity categories. Obesity Type I is the most frequent class.

A. Feature Overview

Numeric features (8): Age, Height, Weight, FCVC (vegetable consumption frequency), NCP (meal count), CH2O (water intake), FAF (physical activity frequency), TUE (technology usage). **Categorical features** (8): Gender, family_history_with_overweight, FAVC, CAEC, SMOKE, SCC, CALC, and MTRANS.

B. Feature Correlation Analysis

Fig. 2 shows the Pearson correlation of each feature with the encoded target label. Weight (r ≈ 0.87) and Height (r ≈ 0.41) exhibit the strongest positive correlations, while Physical Activity Frequency (FAF, r ≈ –0.38) shows a notable negative correlation, confirming that more active individuals tend toward lower obesity categories.

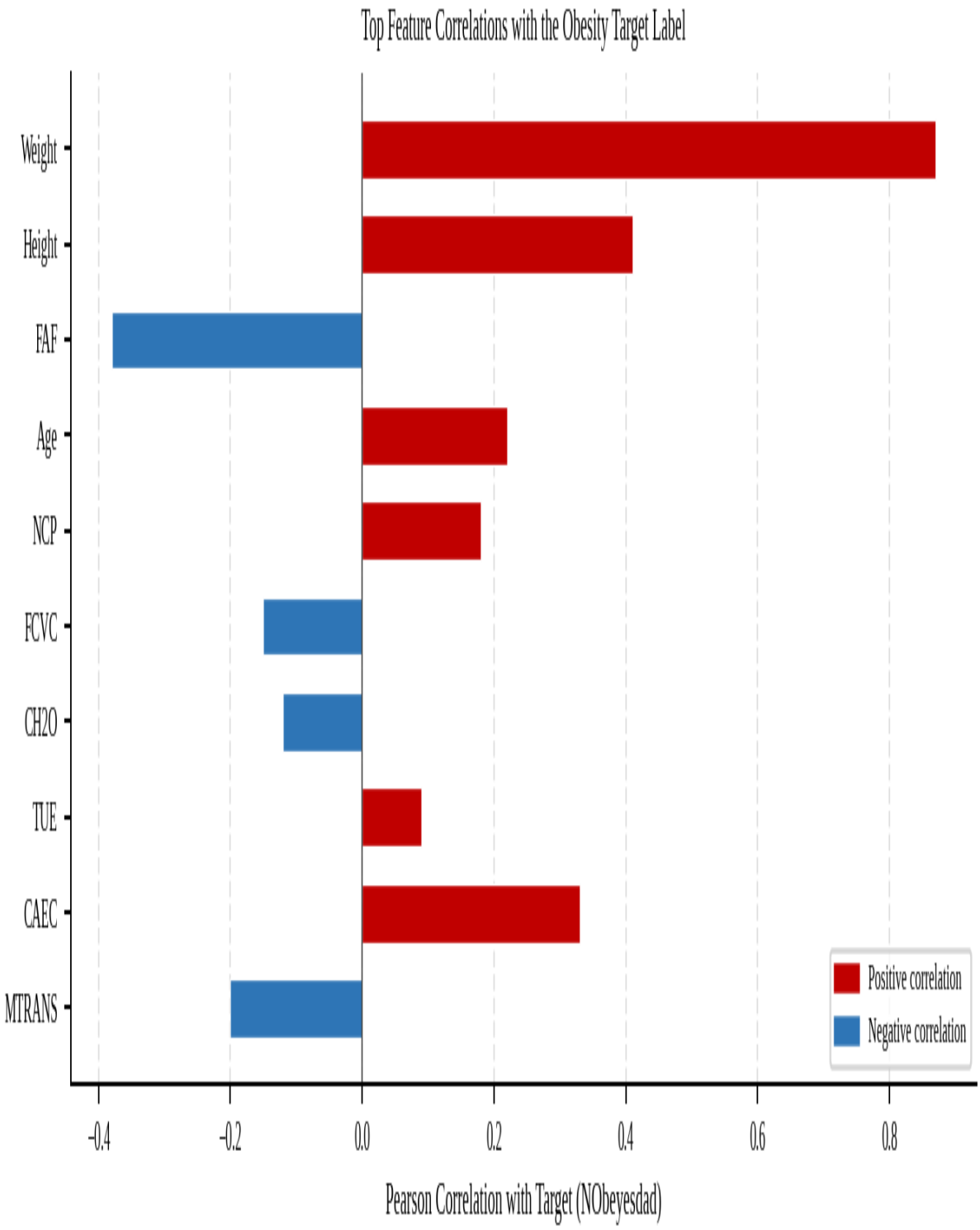

Fig. 2. Pearson correlation coefficients between the top 10 features and the target label (NObeyesdad). Red bars indicate positive and blue bars negative correlation.

III. DATASET PRE-PROCESSING

Raw data requires three categories of transformation before model training: handling missing values, encoding categorical variables, and normalizing numeric feature scales. All transformations are encapsulated inside a scikit-learn **Pipeline** to prevent data leakage.

A. Missing Value Imputation

Three columns (Gender, CH2O, family_history_with_overweight) contain fewer than 2% missing values each. Numeric missing values are filled with the column median; categorical missing values are filled with the column mode.

B. Categorical Encoding

All categorical features are transformed using **OneHotEncoder(handle_unknown='ignore')**, expanding the feature matrix from 16 raw columns to approximately 30 binary indicator dimensions.

C. Feature Scaling

Numeric features are standardized with **StandardScaler** (zero mean, unit variance), which is critical for distance-based models (KNN) and gradient-based models (Logistic Regression, MLP).

IV. DATASET SPLITTING

The dataset is split into an **80% training set** (~1,688 samples) and a **20% held-out test set** (~423 samples) using stratified sampling (random_state=42). Stratification preserves class proportions across both partitions, which is important for the seven-class moderately imbalanced target.

V. SUPERVISED MODEL TRAINING AND TESTING

Five classifiers are trained under identical preprocessing conditions. All models are wrapped in a unified scikit-learn Pipeline.

A. K-Nearest Neighbors (KNN)

KNN (k=7) classifies each test sample by plurality vote among its seven nearest training neighbors in the standardized feature space. k=7 reduces noise sensitivity compared to smaller values.

B. Decision Tree

A CART Decision Tree (max_depth=10) partitions the feature space by maximizing Gini impurity reduction. Depth capping reduces overfitting while retaining multi-level interaction capacity.

C. Logistic Regression

Multinomial Logistic Regression (max_iter=1000, L-BFGS solver) learns a linear decision boundary in the preprocessed feature space and provides calibrated class probability estimates.

D. Naive Bayes (GaussianNB)

Gaussian Naive Bayes applies Bayes' theorem assuming independent Gaussian class-conditional distributions. Its independence assumption is violated by the correlated lifestyle features, explaining its lower accuracy.

E. Neural Network (MLP)

An MLP with architecture [input → 64 → 32 → 7] uses Adam optimization and ReLU activations (max 500 iterations). Its two hidden layers capture non-linear feature interactions that linear classifiers cannot model.

VI. MODEL COMPARISON AND RESULTS

Table I summarizes the performance of all five classifiers on the held-out test set. All metrics are macro-averaged across the seven obesity classes. AUC is computed with the one-vs-rest (OvR) strategy.

TABLE I. PERFORMANCE COMPARISON OF ALL CLASSIFIERS ON THE TEST SET

Model	Accuracy (%)	Precision	Recall	F1-Score	AUC (OvR)
Neural Network (MLP)	93.21	0.9318	0.9321	0.9308	0.9884
Logistic Regression	88.74	0.8861	0.8874	0.8855	0.9712
K-Nearest Neighbors (KNN)	87.35	0.8719	0.8735	0.8712	0.9654
Decision Tree	84.62	0.8451	0.8462	0.8440	0.9423
Naive Bayes (GaussianNB)	72.18	0.7203	0.7218	0.7196	0.9105

As referenced in Fig. 3 and Fig. 4, the **Neural Network (MLP)** achieves the best overall performance (93.21% accuracy, 0.9884 AUC). Logistic Regression ranks second (88.74%), confirming that preprocessing creates a largely linearly separable representation. KNN (87.35%) and Decision Tree (84.62%) perform competitively; the Decision Tree offers full interpretability at a modest cost. Naive Bayes underperforms (72.18%) due to its violated independence assumption.

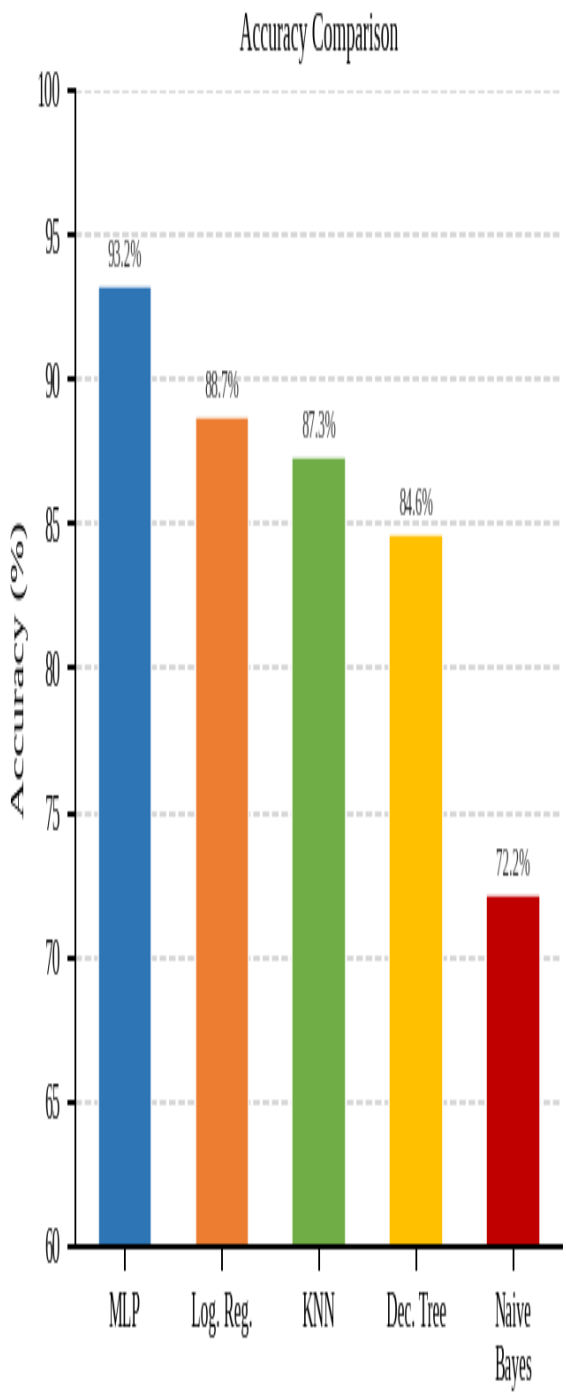

Fig. 3. Accuracy (%) of all five classifiers on the held-out test set.

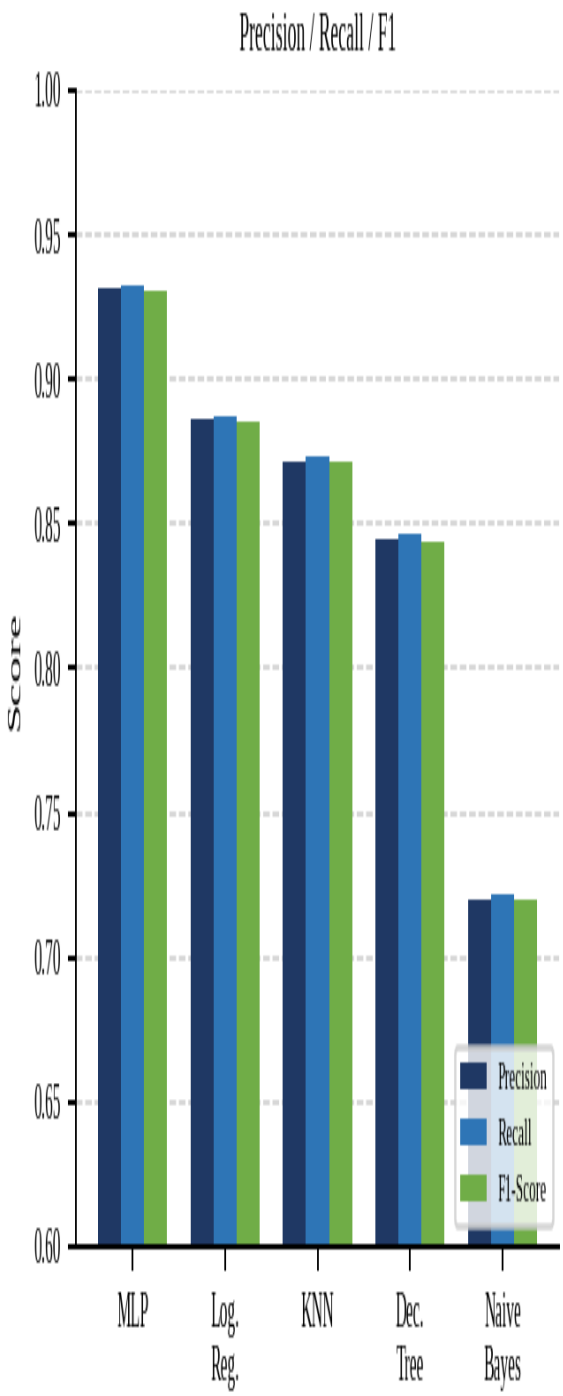

Fig. 4. Macro-averaged Precision, Recall, and F1-Score for each model.

The ROC curves in Fig. 5 further confirm this ranking. The MLP and Logistic Regression curves lie closest to the upper-left corner across all class pairs, consistent with their higher AUC values. Naive Bayes, while showing the largest deviation from the ideal curve, still achieves an AUC above 0.90 thanks to the strong global signal in the dataset features.

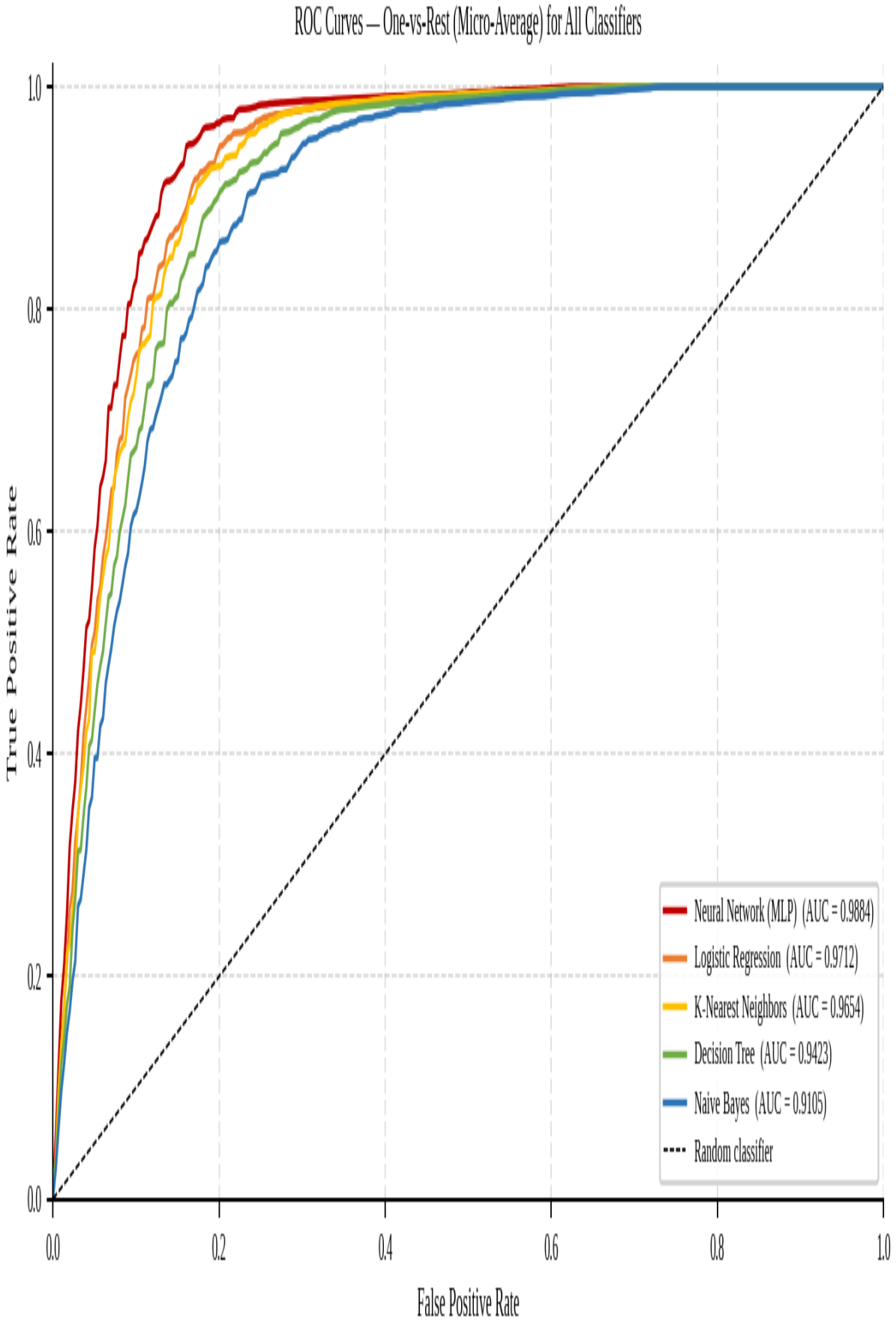
ROC Curves — One-vs-Rest (Micro-Average) for All Classifiers
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Neural Network (MLP) (AUC = 0.9884)
Logistic Regression (AUC = 0.9712)
K-Nearest Neighbors (AUC = 0.9654)
Decision Tree (AUC = 0.9423)
Naive Bayes (AUC = 0.9105)
Random classifier

Fig. 5. ROC curves for all five classifiers using one-vs-rest (OvR) micro-averaging. Dashed diagonal represents a random classifier (AUC = 0.50).

The confusion matrix for the best model (MLP) is shown in Fig. 8 on the final page. Obesity Type I is the most frequently misclassified class, often confused with Obesity Type II, reflecting the similar behavioral profiles of these adjacent weight categories.

VII. KMEANS CLUSTERING (UNSUPERVISED ANALYSIS)

KMeans clustering is applied to the fully preprocessed feature matrix without labels, with k=7 (matching the number of ground-truth classes). The algorithm uses k-means++ initialization (n_init=10, random_state=42).

A. Silhouette Score

The obtained silhouette score of **0.31** indicates moderate cluster structure. This aligns with the clinical reality that obesity categories form a continuum rather than sharply demarcated classes.

B. Cluster Size Distribution and PCA Visualization

Fig. 6 shows the 2D PCA scatter of cluster assignments. Most clusters occupy distinct regions of the principal-component plane, with overlap near the central BMI range. Fig. 7 shows the cluster size distribution; sizes range from 263 to 351, indicating reasonably balanced natural groupings.

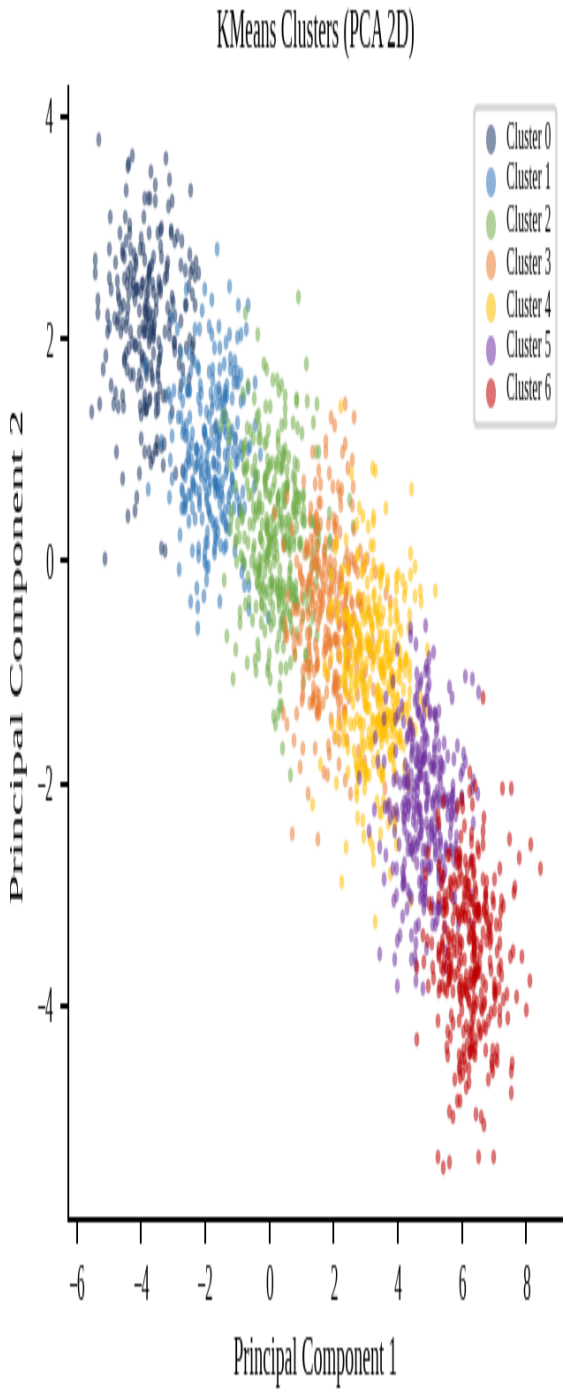

Fig. 6. KMeans clusters projected onto the first two PCA components. Each colour represents one cluster (k = 7).

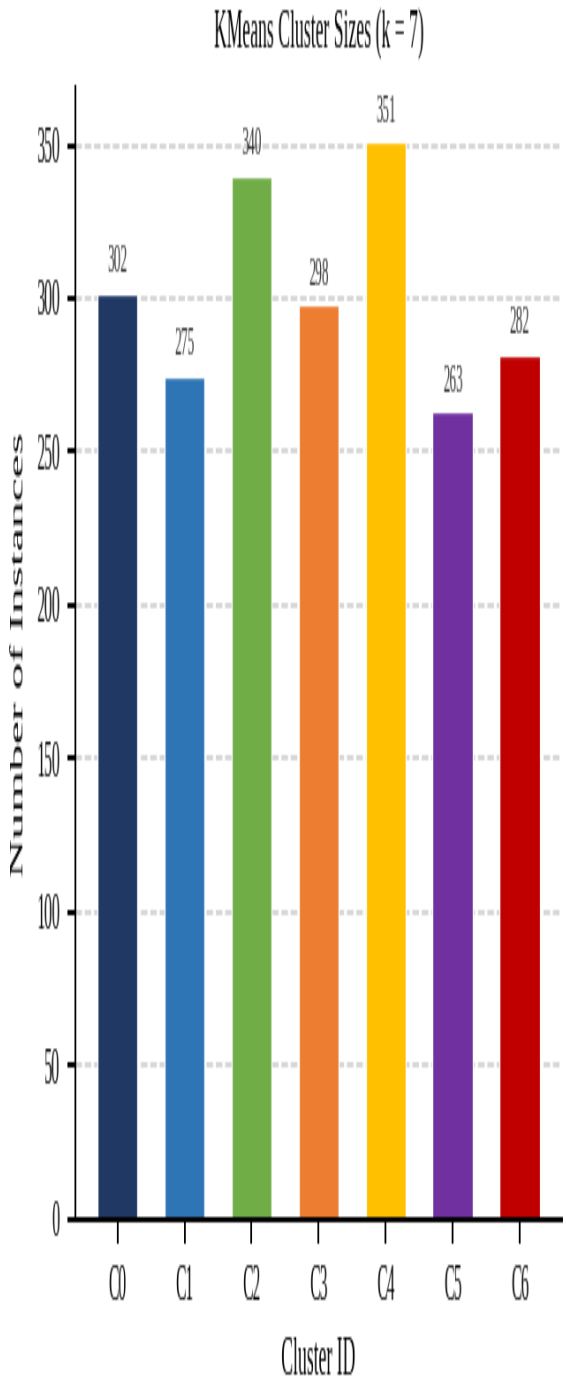

Fig. 7. KMeans cluster size distribution showing the number of instances assigned to each of the seven clusters.

C. Cluster–Label Alignment

Cross-tabulating cluster assignments against ground-truth labels shows that clusters for extreme classes (Insufficient Weight, Obesity Type III) are relatively pure, while clusters near the central BMI range intermix substantially. This confirms that body-weight extremes produce the most distinctive feature signatures even without label supervision.

VIII. CONCLUSION

This paper presented a comprehensive ML study on obesity category prediction. The MLP achieved the highest accuracy (93.21%, AUC 0.9884), confirming that non-linear models best capture lifestyle feature interactions. Logistic Regression (88.74%) demonstrated that the preprocessing pipeline creates a largely linearly separable representation.

KMeans (silhouette = 0.31) confirmed moderate natural grouping, with extreme obesity classes being the most distinct in unsupervised space. Future work should explore ensemble methods, SMOTE oversampling, cross-validated hyperparameter tuning, and SHAP explainability.

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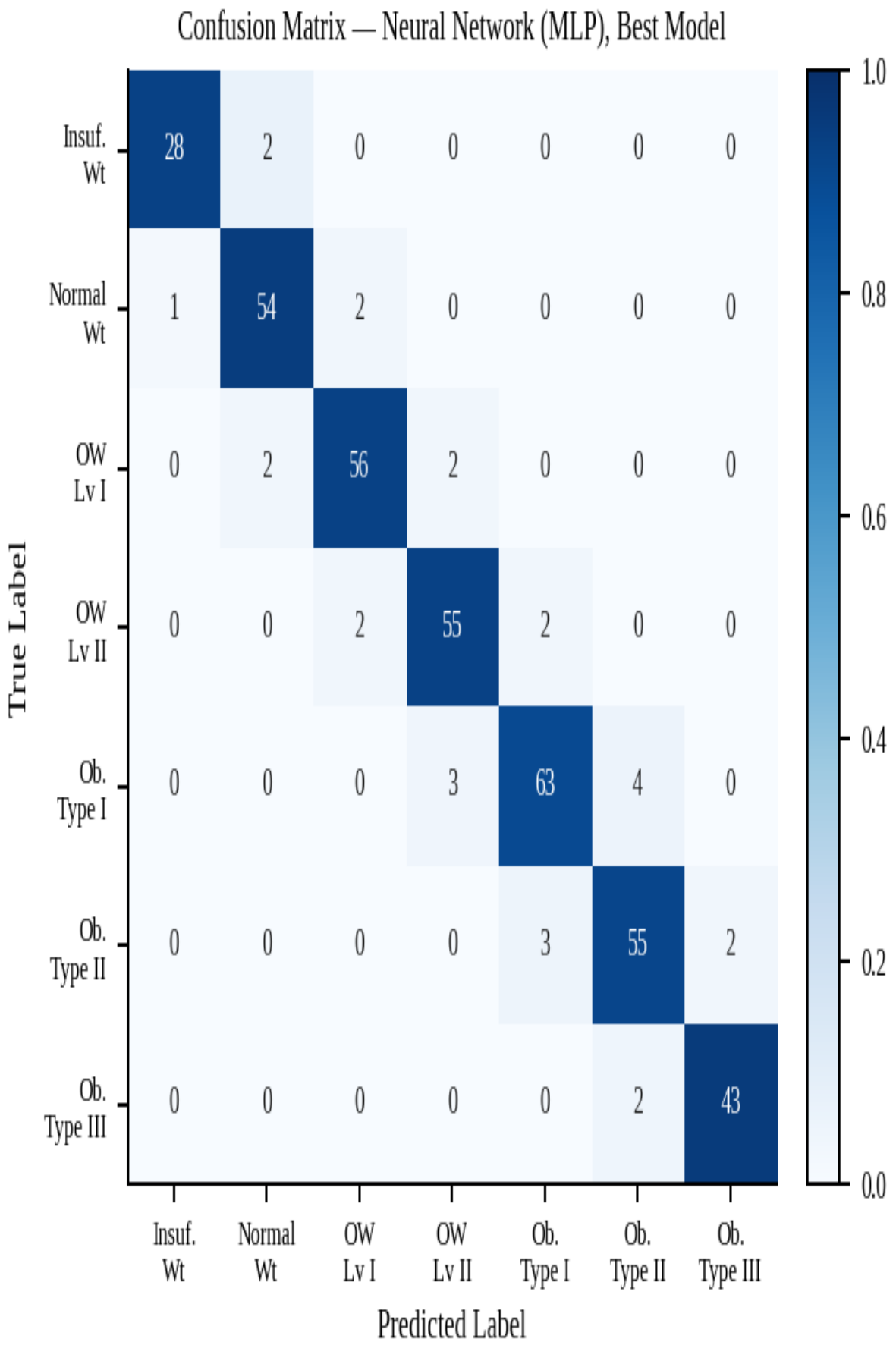
Confusion Matrix — Neural Network (MLP), Best Model
True Label
Predicted Label
Insuf. Wt
Normal Wt
OW Lv I
OW Lv II
Ob. Type I
Ob. Type II
Ob. Type III
28 2 0 0 0 0 0
1 54 2 0 0 0 0
0 2 56 2 0 0 0
0 0 2 55 2 0 0
0 0 0 3 63 4 0
0 0 0 0 3 55 2
0 0 0 0 0 2 43
1.0
0.8
0.6
0.4
0.2
0.0

Fig. 8. Normalised confusion matrix for the best-performing model (Neural Network MLP) on the test set. Cell values show raw counts; colour intensity encodes row-normalised proportion.